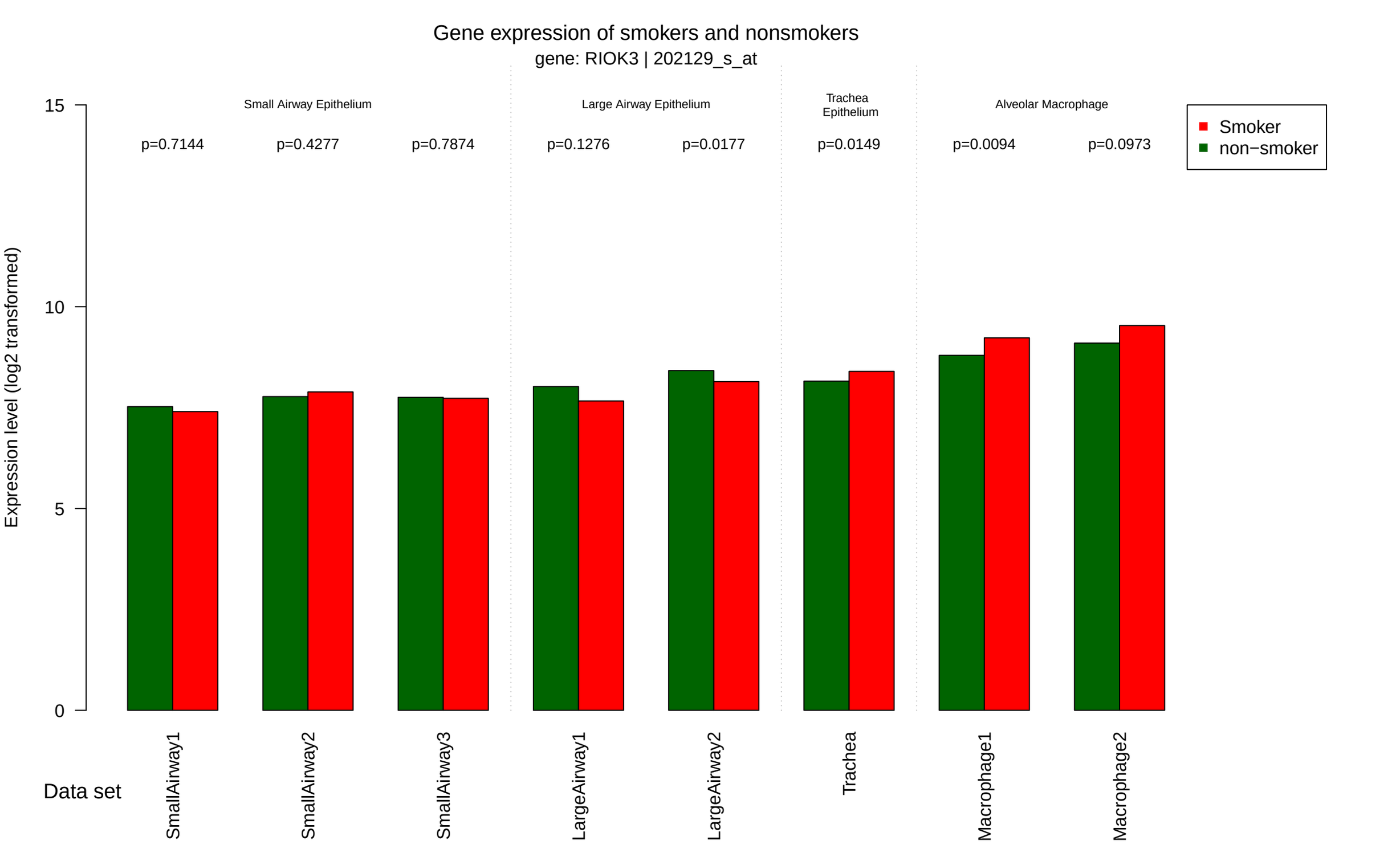
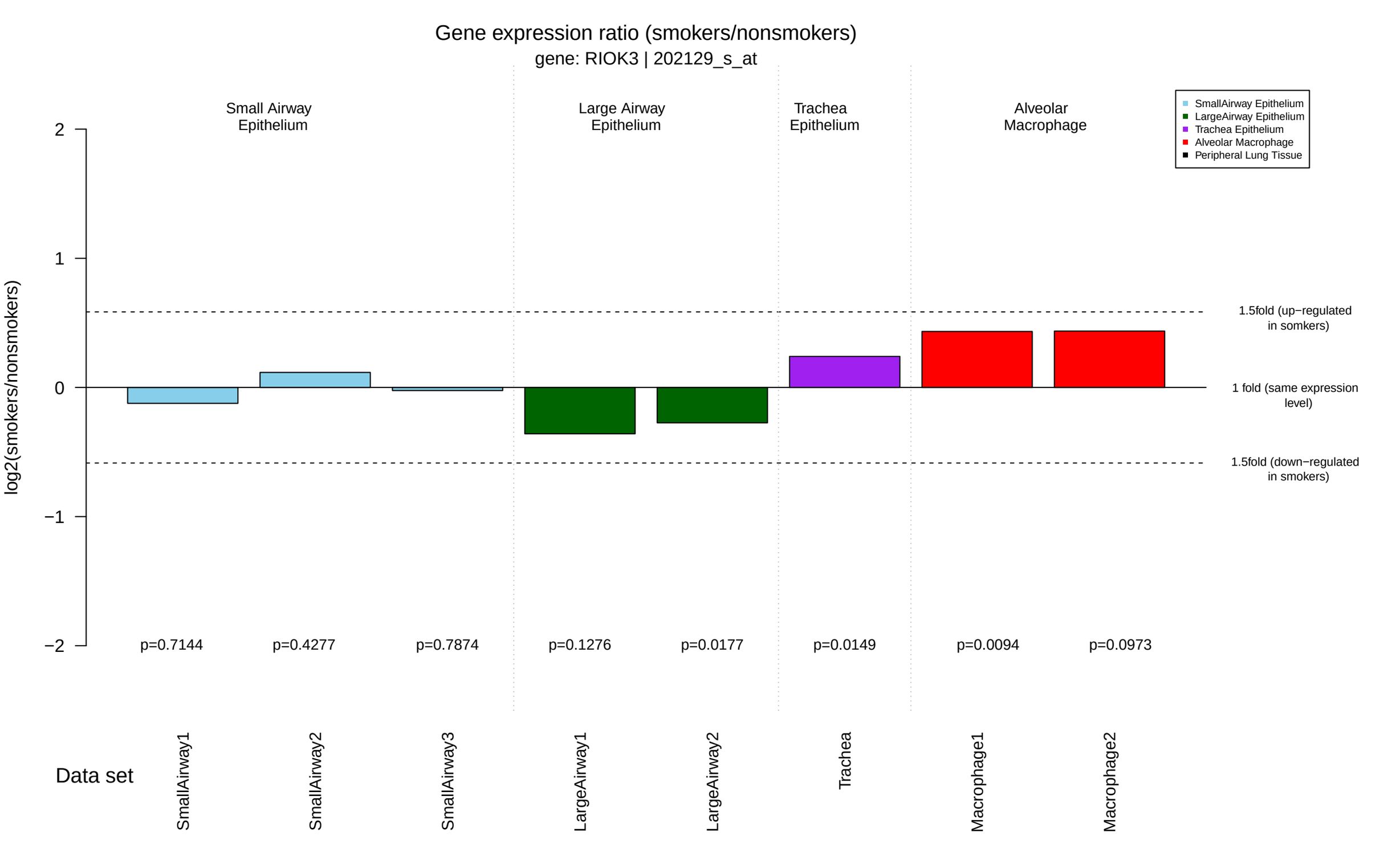
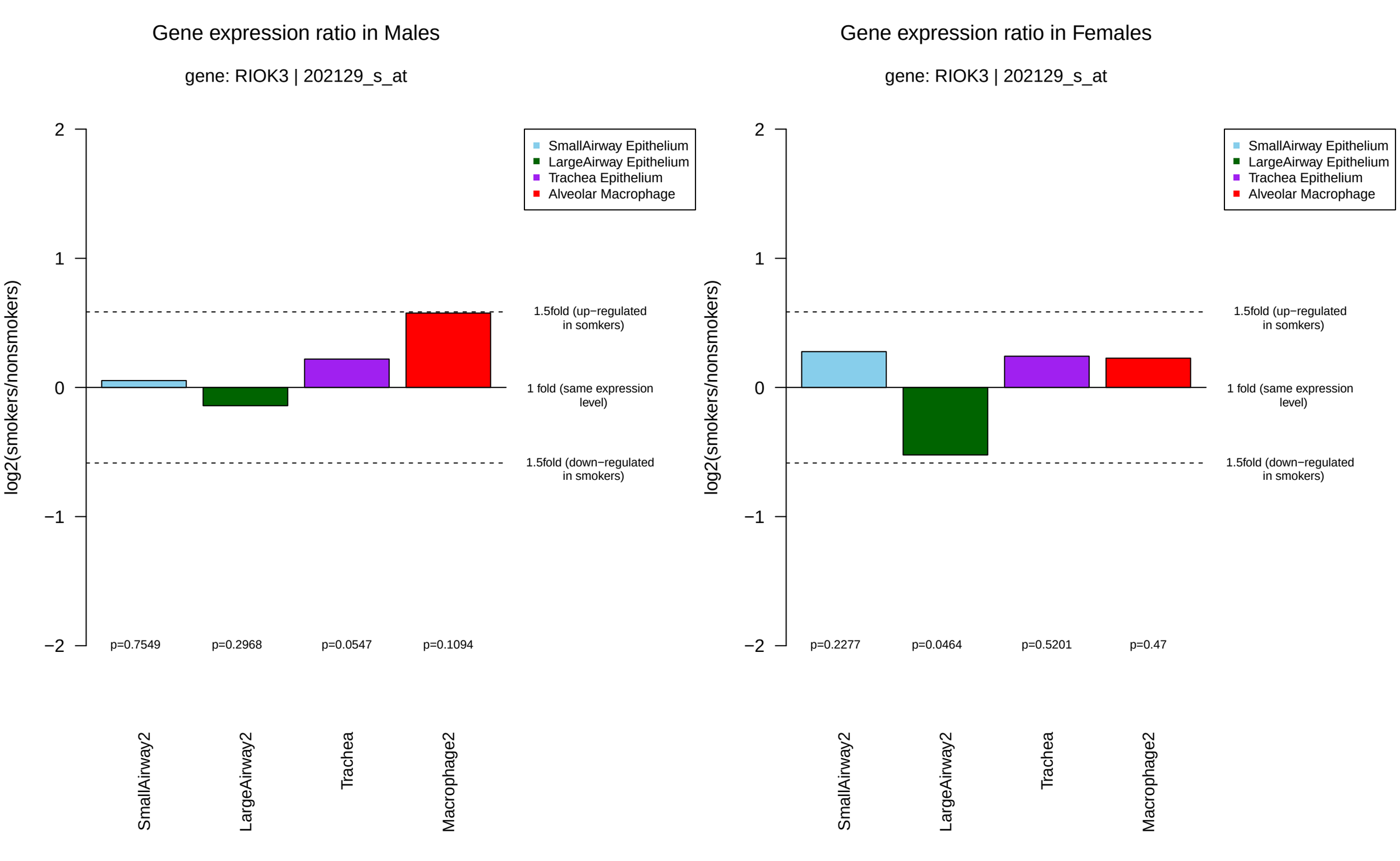
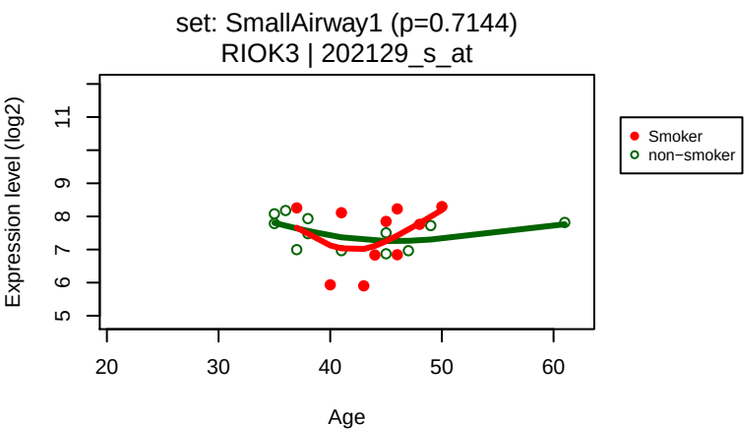
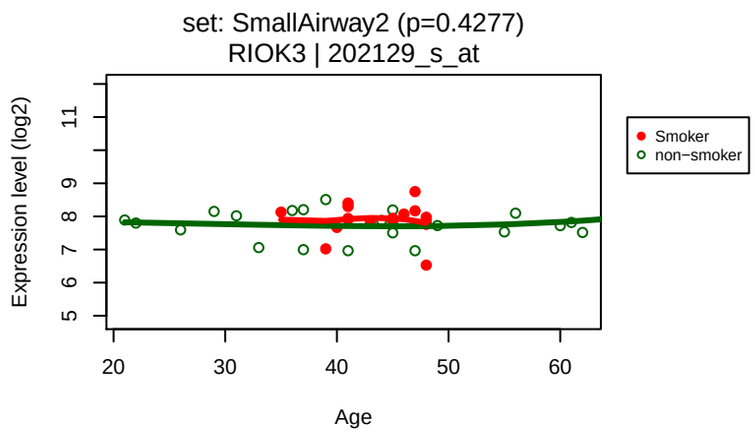
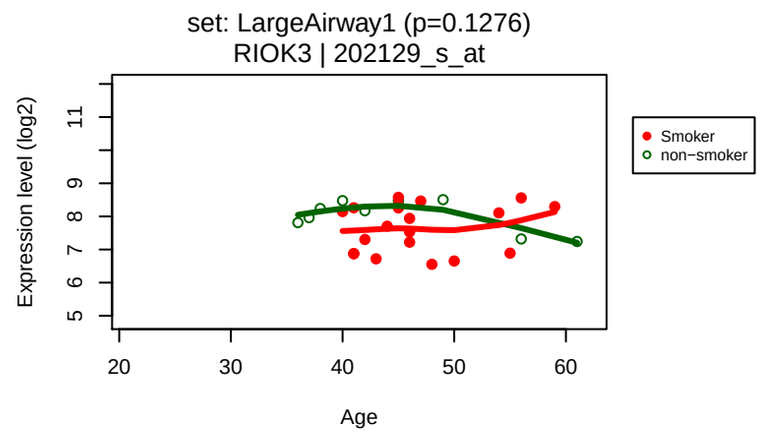

Gene expression of smokers and nonsmokers
gene: RIOK3 | 202129_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7144
p=0.4277
p=0.7874
p=0.1276
p=0.0177
p=0.0149
p=0.0094
p=0.0973
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

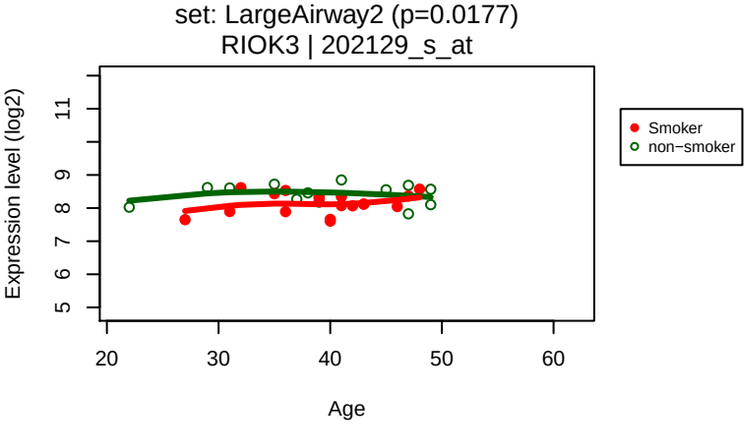

Gene expression ratio (smokers/nonsmokers)
gene: RIOK3 | 202129_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7144
p=0.4277
p=0.7874
p=0.1276
p=0.0177
p=0.0149
p=0.0094
p=0.0973
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

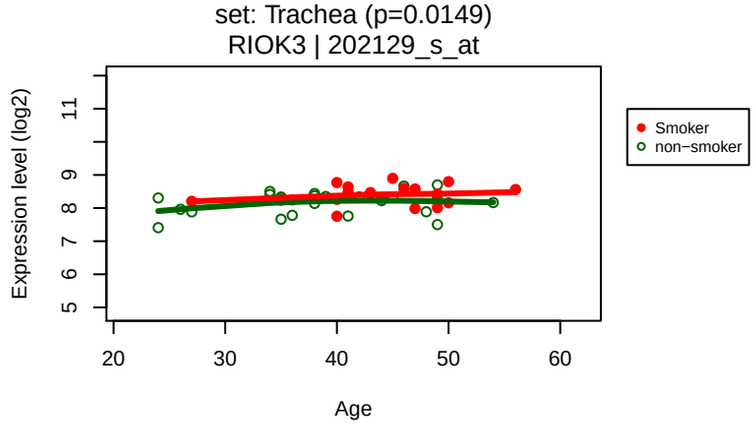

Gene expression ratio in Males
gene: RIOK3 | 202129_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7549
p=0.2968
p=0.0547
p=0.1094
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RIOK3 | 202129_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2277
p=0.0464
p=0.5201
p=0.47
SmallAirway2
LargeAirway2
Trachea
Macrophage2

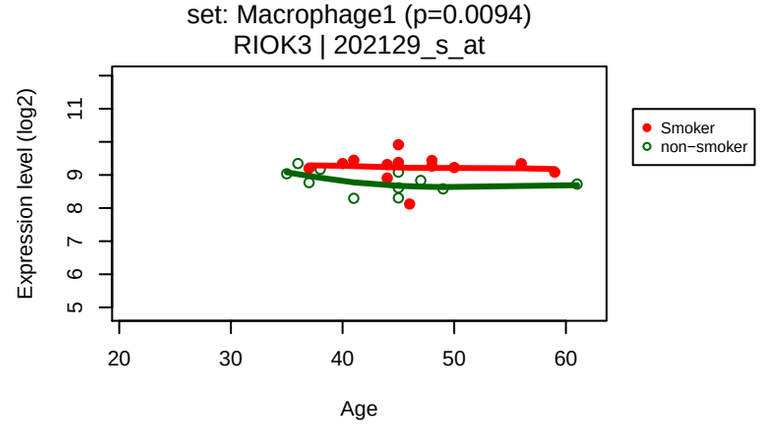

set: SmallAirway1 (p=0.7144)
RIOK3 | 202129_s_at
Expression level (log2)
Age
Smoker
non−smoker

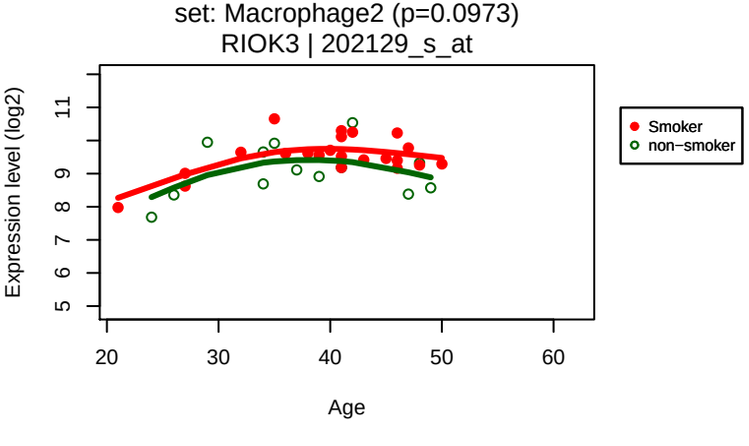

set: SmallAirway2 (p=0.4277)
RIOK3 | 202129_s_at
Expression level (log2)
Age
Smoker
non−smoker

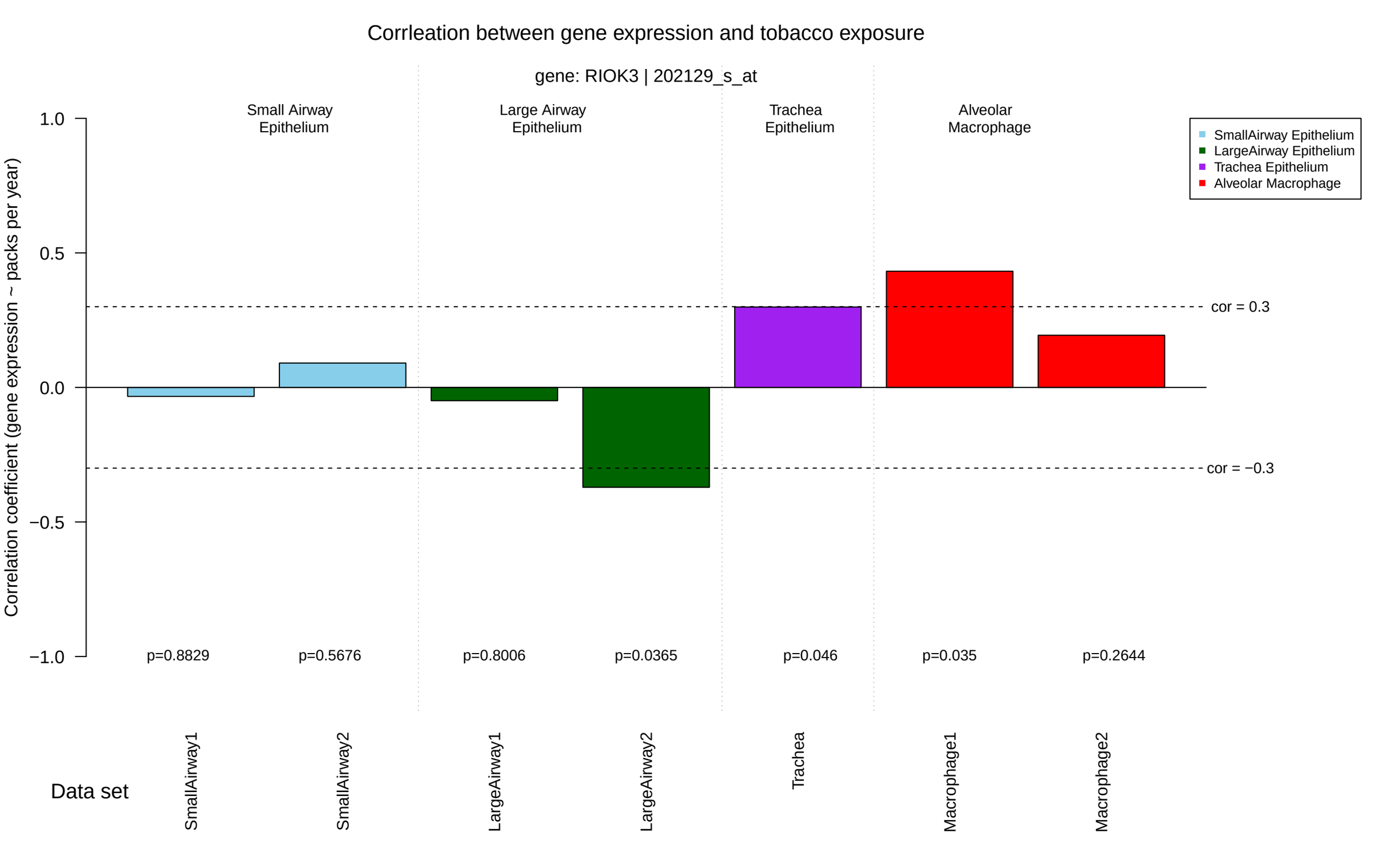

set: LargeAirway1 (p=0.1276)
RIOK3 | 202129_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0177)
RIOK3 | 202129_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0149)
RIOK3 | 202129_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0094)
RIOK3 | 202129_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0973)
RIOK3 | 202129_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RIOK3 | 202129_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8829
p=0.5676
p=0.8006
p=0.0365
p=0.046
p=0.035
p=0.2644
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2